

Pulse Store Operations Intelligence Architecture Decision Compendium

Meridian Group | Pulse Store Operations Intelligence | Synthetic enterprise project artifact

Executive Context

Pulse Store Operations Intelligence is a retail operations transformation initiative at Meridian Group. The goal is to create a unified operational intelligence and alerting layer for regional stores. The program is designed as an operating-model change, not just a technical implementation.

The project team includes Regional Operations, Store Managers, Data Platform, QA, Change Management. Decision records, meeting transcripts and working documents intentionally overlap so a long-term memory system can preserve both current truth and historical rationale.

Business Requirements

The solution must improve consistency, traceability and decision quality without creating hidden systems of record. Key first-phase principles are: No employee-level scoring in phase 1; Enterprise KPI definitions win over legacy regional scorecards; Midwest region is the first production pilot.

Users need understandable status, accountable ownership, explicit exception paths, and enough provenance to explain why a decision or result exists. Material changes must be reflected in acceptance criteria and operating documentation.

Architecture and Dependencies

The solution is composed of Store Systems, Ingestion Layer, Shared Data Platform, KPI Registry, Rules Engine, Ops Console. Component boundaries exist so source-system authority, project-owned processing and downstream user experiences remain explicit.

Cross-project dependencies include ORBIT: shared analytical platform capacity and batch windows. Shared dependency timing is managed through enterprise architecture and project steering rather than by silently duplicating shared capabilities.

Delivery and Risk

Primary risks include missing telemetry, high false positives, regional KPI conflicts, platform capacity contention, store adoption resistance. These risks are treated as business and operating-model concerns in addition to technical concerns because each can change adoption, support effort or the reliability of business decisions.

A dependency delay does not automatically move the project milestone. The project manager first evaluates resequencing, degraded-mode operation, scope protection and temporary controls before seeking a formal decision.

Governance and Traceability

Material decisions retain rationale, alternatives, owner and supersession history. Older assumptions are preserved because they explain why the project evolved. Current truth should be determined from accepted decision status and provenance, not simply the newest timestamp.

Temporary exceptions require an owner and review condition. Shared SMEs resolve enterprise-standard questions; local teams may not redefine shared semantics purely to unblock a milestone.

Decision Summaries

1. Alert scope: Phase 1 alerts target store operations, not employee-level scoring. Rationale: Avoid worker-surveillance concerns and keep the initial product focused on operational conditions. 2. Latency: 15-minute analytical latency is accepted; real-time streaming is deferred. Rationale: The business can act within a 15-minute window and real-time infrastructure would add cost and operational complexity. 3. Pilot: Midwest region is the first production pilot. Rationale: The region has representative store formats, strong operational sponsorship, and sufficiently stable telemetry. 4. Data platform: Pulse uses the shared analytical platform governed by Samir Patel. Rationale: Reusing the enterprise platform avoids a second analytics stack and aligns data-quality controls. 5. KPI governance: Enterprise KPI definitions override legacy regional scorecard definitions. Rationale: Comparable metrics are necessary for cross-region reporting and consistent alert behavior.

Rejected Patterns

Avoid project-local master data, undocumented exception handling, hidden permission bypasses, premature expansion of scope and deleting superseded decisions. These patterns reduce traceability and transfer cost to support teams.